

ActInf Textbook Group ~ Cohort 1 ~ Meeting 19

(Chapter 8, part 1)

Source: [YouTube](#) Playlist: ActInf Textbook Group ~ Cohort 1 Duration: 57:33 |

Uploaded: Unknown Transcript method: auto_caption

hello everyone it's October 7th 2022 we are in meeting 19 for cohort one and we're discussing chapter eight so there's a bunch of questions and also a lot of things to talk about let us just begin with looking at chapter eight does anybody want to uh give any first thoughts on active inference and continuous time everything flows nothing stands still Etc yes um I not on that specifically okay but I will have something to say about this chapter I I worked on this topic um during my my graduate time so what should we get to about 8.3 I'll weigh in excellent any comments like discrete continuous time I know this is something we'll touch on again and again it's kind of like another low road High Road duality with continuous Indiscreet time and I think the textbook tries to give equal time and highlight the hybridization of these models in a way where realistically in the literature most are pure discrete models and continuous time models um have been almost like domain bound certain areas like motor reflexes where there's more of a tradition of continuous models are heavily represented in them whereas decision making models tend to use the discrete formalisms and that difference between the sort of like continuous time traditionally motor Associated and discrete categorical traditionally decision oriented models was addressed in the um folk psychology paper and live stream with Ryan Smith at all okay um also I just started kind of an overview got up through eight three of just understanding and just trying to lay out chapter eight so for any chapter everyone's always welcome [Music] to just make summaries and add their own notes so we can look at the questions many of which are on the earlier part of the chapter and so maybe we can look at the earlier part of the chapter more this week and of course then have later time uh okay so 8.1 it's a single paragraph it's a complement to chapter seven which was a discrete time approach towards generative model construction the focus here is on continuous State space models which are well suited for physical fluctuations sensor receptors and continuous motion of effectors they start with the case of movement control 8.2 let's get familiar with the notation the states hidden states are going to be X the data the observables are going to be y and then

how States evolve over time depends on a static variable V so that could be a hidden or slower changing cause or factor in the environment then we have the Omega terms associated with each of the data and the hidden States change Through Time and that variability u_m is like the noise and this is like the signal or the flow so it's kind of the lengthen decomposition or language approach towards dynamical systems modeling one interesting thing to note is first that this is hugely foresaged in u_m SPM then u_m also that the data are a function of the states at that time whereas the equation for the hidden States is a change in Hidden States $\dot{x}$ note that action is absent from equation 8.1 this is because action is part of the generative process not the generative model thought that was kind of an interesting question what does that mean that action is part of the generative process and not the generative model surely action selection is part of the generative the cognitive process of policy inference active States are part of the particular States active states are blanket States particular states are the generative the generative process is referring to the external States the hidden States so is action part of both the generative process and the generative model because it kind of sits in between intermediates or is this referring to the consequences the generative model only deals with those variables directly influenced by States external to a Markov blanket so is that saying that the generative model is kind of like the complement of the autonomous States action internal States or is this even more extreme it's only dealing with the variables directly influenced by States external to a markup blanket so then it must only deal with sense States which is like a narrow perceptual control reading and in that situ in that sense action execution is outside of the generative model because the generative model is only dealing with observations does anyone have thoughts on how generative model action and generative process are being used Eric I wonder if you can consider your own actions part of what gets observed under that framework as far as I would uh the Apologetics for the narrow View would be exactly that which is like you have um uh proprioceptors in your elbow so you don't observe action but you are getting continuous updating of perceptions that are extremely influenced by action but the hidden state of the world is actually like the angle of your elbow and so you're not directly observing the hidden state of the world nor how it changes all you have is the proprioceptors and of course you engage in policies to change your proprioceptors bring them towards alignment which is what we're going to be discussing in the chapter Ali I saw your hand race uh yeah I wanted to mention exactly the proprioception which you mentioned so I lowered my hand okay yeah great but then um could you include your you know the fact that you've got signals going to your muscles your your motor signals and motor neurons are in there making

your muscles contract can that be part of your observation to say well okay if this is happening that these muscles are getting activated sure enough the the arm moves an appropriate perception changes so you can say well from from the outside of that box that's all part of the generative of what of seeing you know but not part of generating the those muscle commands which is seeing what the effect of them is I wonder if that would be an acceptable viewpoint cool well we just we'll return to this like where is action what is action but I think it's just important to note while we're early here continuing a little bit on this part I thought was also a little curious impact so this is equation 8.1 we mentioned equation A2 so first question italics versus bold G and F do you think that that matters I don't think it matters does anyone disagree like do you think it's moving from like a some sort from one space to another space or do you think that's just the typographical error I think it can be considered as an errata um one thing I did notice is that 8.3 and 8.4 the italics they call a generative model an 8.4 they call a generative process with without the italics could that be the distinction oh um okay eight eight point three and four yeah ah okay it's a very good question well let's just say italics an 8.1 and 3. old in 8.2 and 8.4 um unclear if relevance difference foreign question I mean if the meta model you say frames it so if you consider the niche to be a meta model then sure the difference between equation one and two other than the Bolding or not is that V the static variables have been replaced with u actions so this takes us from like the sort of SPM where you'd be like you have neuroimaging data y as a function of underlying neural States X and then neural States changing through time are a function of neural States at that time and some conditional Factor that you're testing for like whether they were exposed to treatment water treatment two V so this is still a um partially observable dynamical model but then it gets moved very neatly into the control theoretic and it policy selection is inference Space by just saying yeah those slowly changing factors that influence how the hidden States change we control those or we have some ability to influence there yeah so this could be a cue towards the Bold difference which is like the functions G and F are not necessarily the same as those used to define the generative process if they said not necessarily the same as those used to define the general process bold G and F then it would be unambiguous but it's a little ambiguous we do not need to explicitly write down the Dynamics of action in the generative they emerge from a choices made for terms in equation 8.1 we start with a simple sort of so G which is the function that's generating the data is going to be X f the function of how X changes Through Time is a function of X and V the slower change in causes and it's the difference between those two the hidden State represents the expected value for the data or in

other way say that would be equation three says that the expected value for the data are the hidden state but does the same and it's a point attractor or in the Bayesian mechanics we could say it's mode matching by a tractor uh yeah yes the attractor here is the mode tracking that is being referred to here when there's a difference between the hidden State and the hidden cos V it moves towards it so it's just moving towards a single fixed Point attractor now the generative process is defined so it looks almost identical except these are replaced with u's and this differential tracker this kind of PID control Motif is replaced with action it's an expression of the equilibrium point hypothesis which treats motor control as enacted by reflex arcs that draw limbs towards equilibrium points set by descending motor signals in uh and then also this is related to the skilled performance live stream 23. the scheme does not require specification of inverse models that are widely used in other formulations of motor control 8.5 makes a slightly more sophisticated that's based around not just position because of course you can't just like pick up your finger and move it to a different position really the action affordances are in the space of changes in velocity which is to say acceleration so this is kind of a classical mechanical perspective on motor control and it also points towards uh generalized coordinates of motion and that kind of PID control and the Bayesian mechanics and so on now we get to 8.3 continuous time formulations are well suited to characterization of movements well certain kinds of movements on the chess board maybe not but some kinds of movements sure Eric yeah I like to point out with um equation um just above that where he's got the they've got this um spring Mass that um that system oscillates forever right it's got a spring and a mass and there's no dissipation of energy so it's it's a step up I guess better than um in a more advanced than um equation 8.3 in which you have instantaneous acceleration so they recognize okay you can't have instantaneous acceleration but there's no there's no uh damper here so normally the the the true equilibrium point control formulation is is viewed as a spring Mass damper system and that way if you if you remember from physics class you know depending upon your set your your mass the spring tension and the amount of damping you can get over damping or under damping and then you can get kind of an optimal settling to the uh or uh approach of the set point with maybe just a little bit of overshoot or no overshoot but if you don't have a damper in there then you it just never it never cut cells in your at the equilibrium point but um so just to carry this a bit further um I I have to would have to to disagree with this statement that continuous time formulations are well suited to characterization of movements because that's that's assuming that a the equilibrium point control model is a good characterization of movement I worked on this in grad school for

about two years when I first got to grad school we worked on the equilibrium point control model in I work in the BT Lab at MIT and we we had an apparatus um that mimicked the apparatus that was used in monkeys we had one that we used on ourselves as human subjects which would um allow you to move your arm and then it would kind of give you a little jolt a little a little disruption to measure the amount of stiffness and in your movement as you're moving your arm as a way of trying to deconstruct or or work back the parameters of the equilibrium point control uh model for for movement and it's attractive because um as they say in this in the chapter um well you don't have to have any very sophisticated um knowledge of motor planning or inverse model of the of your your plant the system in order to all you do is you say well this is where I want the arm to go and passively uh it takes care of itself perhaps through um a feedback loop in the you know through the spinal cord which I guess is what 8.1 has um it's interesting that that there's a paper from 2009 uh 11 and Feldman that is you know reviewing this but um our conclusion was pretty much that for motor motor control is just it really is is not a good model it doesn't work in order to make it work you had to have all kinds of play all kinds of shenanigans like well you're not just have a fixed set point you might overshoot and then you bring your set point back and you actually have to control the stiffness during movement in order to account for you know observed data but still it falls short because it because many if not most human movements or any you know motor movements um are ballistic so it's not that you set some conditions and then your your spinal cord and motor and muscles act to fulfill those conditions that you sat once from the top from your brain what really happens is you're you're descending signals anticipate a whole trajectory or a whole plan in advance of how we're going to accelerate and then we're going to have to decelerate and so it's it's the motor planning kind of knows what what your plant is going to do it doesn't wait for the plant to do something and then have to respond to that in kind of a passive way at the lower levels um so it's just like you know if you play the piano you're not waiting for your finger to say am I far enough down on the key you you kind of you learn how hard you have to press in and when you when you start to the release before anything happens I mean it's our movements are very very fast and and um so there's really no getting around having to model the plant and doing doing this sort of inverse modeling that they're they're concerned about um in in order to account for how how any creature really moves or anything you know above an insect and probably an insect who's um undoubtedly of it's a motor plan so um so I really find that this this chapter um is is is really not um I don't think it advances the cause um active inference it doesn't really to me um it doesn't really fit with with with with with how about a

control really has to work and doesn't add anything to the theory of it I mean it's using an old theory that doesn't work thanks for all those points that's my statement that's it that's the tweet um well one uh live stream that other than 49 which has just been like amazing from the physics angle we're preparing also for 50. which is using it's also by felden Barrett and others and it's specifically on interceptive reflexes so less exploring like the spinal reflex arc and more on the vasopressive interception but using those kinds of still classical very differential equation based equilibria control and showing that those types of formalisms can recapitulate like allostatic Behavior it's always difficult to know like is that a very narrow and fragile range of parameter space is this a descriptive model that's making unique predictions let's let's look at figure eight one because it's speaking to um generative model generative process distinction that was mentioned earlier and it was addressed in a question how is this representation of the reflex arc similar or different to what was being used in figure five or in chapter five so in chapter five here was the summary statistic or summary graphical abstract there was a lot more focus on this kind of like um policy selection and so on but we we um did have this kind of a reflex arc why e and so on okay so now we're in 8.1 is the action selection component so here is that simple attractor being shown where it's just X is going to be converging to V and this is the change in external States so here we have μ of x in an equation in the lower one is this one referring to this or is that kind of bleeding out from the bottom panel because here's U of X and here's you have μ of x here it looks like μ of X is a prediction modulator on a differential so it's like if the observation is where you want to be then there shouldn't be any action but as the observation diverges from where you want to be you have a Precision modulated action is that active inference or is that just a notation um a different notational representation classical equilibrium point motor control theory do those models integrate sensory and action together I mean in one sense they definitely do nobody was ever generating reflex arc models without some sensory feedback variable so that's open important question how is this is this didactic with respect to active inference now just kind of saying yeah one special case of active inference is um the traditionally understood and limited equilibrium point model of motor control or is it like this is our new way of doing Motor Point control okay just kind of continuing on all right Precision attention and sensory attenuation so here's another curious note so earlier we we uh discussed whether motor control was or wasn't a natural setting for continuous in the Box it won they write we addressed this the importance of precision in chapter seven but it's worth recapping its role in continuous time systems in many ways this concept of precision is more naturally addressed in this setting as the π variable

appears a direct consequence of Laplace approximation but can't you do a LaPlace approximation in discrete systems as well LaPlace approximation is just mode determination followed by the parameterization of a parabola to create like a gaussian or at least a concave optimizable two parameter approximation to an arbitrary distribution I don't exactly see or is it referring to the LaPlace approximation is itself a continuous function but you could apply a LaPlace approximation to a categorical distribution as well so in what ways is precision more naturally addressed in a continuous model is it that Precision is a continuous variable like Precision isn't an integer valued we can just leave that one okay but so we went into eight three dynamical systems kind of looked at the spinal cord again but it may or may not just be a sort of like trivialized or didactic representation of extremely traditional representations of motor control then box one Precision attenuation attention sensory attenuation all right now we get into the lot kevaltera Dynamics what are some formal and qualitative implications of the use of Lotka Volterra Dynamics and Lorenz systems so I think there's a few interesting angles that that us here now but others just more um broadly may be interested in so one is lockabletera ties in of course with ecology and population ecologies and so that is also very interesting to consider in terms of like winnerless competitions and neurodynamics and that is basically the analogy which is like just like even if you have a Negative Edge between the predator and the prey for any persistent which is to say oscillatory or manifold attracting population at some point like you have to let the prey regrow so the analogy in Neuroscience that's fleshed out in SPM is like if you have a brain region even with a unilaterally negative influence on another region at some point it has to like let off the break cyclically so lockabletera summarizes these kinds of like winnerless dynamics that can give oscillation and time structure including with like complex Dynamics and recurrent Dynamics limit attractors all these like kinds of dynamical processes can arise from a pretty simple starting point and uh here we see manifolds and like limits on the projection because it's a three-dimensional space plant herbivore carnivore or that could be the um blood oxygenation of three brain regions in SPM and this is their neural trace or their inferred activities and so then you can find like manifolds of lower dimensional activity and then also one um kind of related note is the Volterra series and this is a pretty nice just technical note which is Taylor series does expansion arbitrary functions from a starting point and the Volterra series is able to include like a window of inputs also known as the memory effect and so it's discussed in um SPM it has a continuous time and a discrete time variant and it is just a kind of Kernel estimator that can be used for dynamical processes input output dynamical processes with memory so it's

kind of a classic in that area and I think even in some live stream maybe 45 I asked Tristan about the lockable terror models and about like like where did we move away from them or something like that but that'd be a good answer to kind of come back to all right so um now those kinds of lockable Terror Dynamics are going to be applied monochromatically as ever to active inference and that was one question which is uh it didn't finish writing how is the lockableterra model being used in the case of handwriting and herrera's paper which I just brought in but I didn't really investigate deeply the left there's two different applications here left eye blink conditioning so here it looks like a conditioned stimulus unconditioned stimulus where there's like where something happens and you blink as we've mentioned previously like it's a little bit difficult to tell exactly what these expectations are of or anything like that but this is like some kind of stimulus response blink blink once if no Twice of yes once if I land twice if I see and then on the right side sequential Peaks using an attracting but selecting the specific attractor on the basis of which population a lockableterra system is currently highest this leads to sequential visiting of each points giving rise to a type of handwriting so this is kind of interesting and I I know for Jacob and Ali also with our mode tracking path matching discussion Bayesian physics um so it's like um I um I must it's Bart Simpson at the chalkboard this is I must rewrite the sentence I must rewrite the sentence I must so your expectations are that you're going to be writing this sentence and then depending on which of those expectations are on top that is going to set the motor imperative so it'd be like if you were writing the letter um D and it'd be like well it's like first there's like an arc down and to the right and then there's an arc down to the left and then it's a straight line up and then it's an arc down to the right and then it's an arc down to the left and then it's a straight line up and so here's kind of like doing this writing as it's picking a mode pursuing the mode and I think it's fair to say uh jakov early please add any more if you feel like it but like shaking off the slumber of the classical physics and still having one foot in both worlds and the three faces of Bayesian mechanics are a much more advanced way to discuss this issue yeah Jacob what do you think I was just going to say that this well not the case with writing specifically but uh the case of motor uh motor control was discussed in Thomas parr's lecture um at the CPC 2022 which I don't know if it was recorded uh actually I think it was uh but I don't know if it's public yet um but it's interesting that this example is in the continuous Time chapter because you could think of it as a nested generative model with both discrete and continuous uh uh factors like if you partition it into three decisions like uh for a straight line up then an arc down then a second Arc down um those are three discrete decisions that are at the top of the of

the hierarchical generative model with the finer grain motor control below thanks also just here's that classic three faces figure mode tracking this can be seen as using a log of ultra pseudo random number generator to spit out higher level predictions about which mode should be pursued then as to the Dynamics of how that mode is approached one can either use the equilibrium points methods discussed in this chapter or the more advanced Bayesian physics methods to describe that in terms of gauge Theory and so on is that fair to say oh is this it jakub thank you Ali for sending it uh it's it's not the same it's not the same one but it looks like a lot of the slides are the the same I think the thing that I was referring to I'm just trying to find the slide oh yeah I think it might be minute 41 okay I I really can't wait till there's some open source webcam software to do isicade analysis and um pupil diameter I don't think it would be that intractable maybe you'd have to be kind of close or you'd have to use a 4k camera or something but is there any software like that available I've never come across it just like move the mouse to where I'm looking and then click when I blink twice isn't that kind of like that exists that does exist but it but what is it called or where is it being used yeah I know it's being used uh for um people with physical impairments who can't control the computer yeah otherwise yes I don't I haven't I I haven't seen I haven't seen uh like an analysis of people dilation but eye tracking is definitely a thing okay maybe we can explore it um all right so yeah the blinking example was a puff of air so it was just like a beep and then a puff of air and so then they learn that the they learn to preempt the air Puff learning in continuous models eight two continuous time domain accumulating evidence this works if we treat data in a series of small time intervals oh no don't discretize um this relates I think to some of the finer points on like which stochastic calculus is used like the Edo calculus in the Bayesian mechanics we won't go into it unless anybody really wants to say more but this looks like important just to learn about the integrals and the gradients okay pomdp has largely superseded the use of the generalized lockable terrorist systems inactive inference applications and that was kind of I think related to asking first about why things moved away from the lockable terror formulations Lorenz system also explored in stochastic chaos Markov blankets live stream 32. it's it's almost it's dealt with as minimally as you could possibly deal with the Lorenz system in eight seven okay eight four generalized synchrony so multi-aging inference starts coming into play as well as agent environment synchronization and we actually like talked a little bit about the bird song in the previous hour um now the Run system comes back and yes it's dealt with like sort of in the 2015 frith and friston paper active inference communication and hermeneutics and that's where the synchronization manifold is kind of displayed and then

more recently in live stream 32 is um and this one has a DOT three I think this might be the only dot three as of this point I think Connor Hines could only come at a later date yeah so these are good discussions on the lorenza tractor a couple Duran systems this is kind of um very similar if not identical to the uh bird song synchronization manifold in first and forth 2015. birds birds birds bird synchronization isn't real then hybrid discrete models hybrid mix models not that kind of mixed model allow inferences about sequential action plans and translations of those decisions into movements through a continuous model in that respect it's extremely linked to livestream 46 active inference models do not contradict folk psychology which is all about that continuous time motor active discrete state-space decision active inference Mai and Dai and using those motor and decision active inference while lower level model in the dashed box is the same form as the other models um in in other words it's continuous because we have V instead of Π for the causal factors and F instead of B and G instead of a and uh η instead of G is it some mapping between well in this case I think just looking at it a little more narrowly how um outcomes map to slower causes of the world so here this is a discrete model up top with a G oh yeah that was one era that we found free-floating G can't live with it can't live without it but it should be connected to the π of course there's a discrete time model on top and then it is kind of cascading its outcomes down to these continuous time models so one could think about this like the lock of Volterra handwriting example kind of could it's not the exact structure of this but it's related um this is um something I know Jacob has thought um how discrete and continuous models um interface because we've talked about how like the temperature could be continuous and the thermometer could be continuous but then the internal generative model could be like is it hot or not so another question of how to how to move from generative models within between different parts of the model where where or what kinds of transitions between continuous and discrete State space is possible here's a continuous um isocate uh ocular motor paradigm where the continuous lines are still confusing but at least they are Justified eight three combining categorical and continuous generative models outside of active inference has primarily been framed to the lens of clustering the aim is to assign data points into a discrete cluster sure about that I think clustering is one discretization clustering and thresholding is one discretization approach that's been used also just binning seems like a simpler method how do you say a continuous distribution of one through ten and make it categorical just bin it one through ten or you could cluster and do thresholds so then you could have um cluster one with 9.7 to 9.9 and then 9.7 to 9.5 and then eight six there's a big table here's the songbirds this

author isamura I believe is the one who will join per 51. in November and I think that'll be like our last probably one of our lost paper streams for the year we have some we have just this last discussion with Dalton next week and then interception and then the canonical neural networks perform active inference papers but isomers worked with power and first and others oculomotor reflex eye Pursuit psychosis illusion cicade action observation attention hybrid and self-organization models well with four minutes left kind of an interesting chapter but I can only know it how I'm seeing it and I have serious questions about what it does in the book Eric um one thing I was puzzled by and maybe you guys can help me out with on the bird the bird song examples were they implying that um Bert starts singing another bird picks it up synchronizes using these Dynamics um the proof of that is this other bird will just step in and take over and the first bird can stop I don't believe I've ever ever heard that happening are they suggesting that that happens or or what okay I agree like whether it's a improv whether this is a bird improv conversation in this case they're singing from the same hem sheets so it's like somebody should be singing the bird anthem and now we're going to be engaging in this sort of Quasi dialogue where we're going to be reducing our surprise about hearing the song that we expect to hear which goes like you know insert your favorite song there and then the phenomena that gets highlighted is the way in which um one bird is singing and then as it's singing its Precision starts to drift it stops singing and then the other bird finds itself picking up at the same to reduce its surprise about that song being sung at the time I think especially with it with a bold title like land claim to hermeneutics more broadly singing from the same pre-coordinated hem sheet is not the same thing as information updating or even just pure expressivity so I I think it it leaves gaping areas for people to actually develop because it's like well we have what don't we have databases of like millions of bird songs and natural soundscapes so like we need some bird watching active inference practitioners who want to actually develop the bird song Motif because otherwise it is too easy and ironically becomes singing from the same hymn sheet oh well hermeneutics has been addressed in 2015 synchronization manifold 2015. whereas this is like a very Fringe case of communication that is based upon most charitably High reliability execution of pre-ordained performance which is a tiny Edge case of communication you know my understanding of bird songs is you know the the um baby birds here here the with their parents saying you know they've got some predisposition to to sing a certain kind of song and um for most birds you know there are some that are very adaptable you know parents parrots you know stuff like that but most birds um learn the accent and the or the um the dialect that they are raised in and that's the song that

they always sing um and there's some variation on it I mean maybe they'll stop Midway through but what what are the true facts about I what I saw to be a claim here where birds will pick up from each other or one leaves off does that really happen I I I'm not familiar with that happening I've never I don't think I've ever heard it happening and what my impression the way most birds sing is they'll sing their song all the way through and there might be a column response another one might pick up uh you know might answer but I don't think I've ever here synchronizing of songs um but I might be just not well enough informed about the Ecology of those things yeah very interesting um and the whole area with game theory of song it's beneficial that somebody warns about the Predator but then you've revealed your location when you sing and things like that like that's the kind of decision making and um uncertainty resolution and risk evaluation that I'd imagine in ancient-based bird song model wood oriented towards but those are some cool ways for us to go so um next week I I guess this is kind of a shorter chapter and it has some Arcane symbolism but also seemingly very skippable symbolism so I'm glad that we could get through the whole thing and um will return to any further thoughts on eights and then maybe we can have more of a uh seven plus eight synthesis as or anyone can look into any of these citations key advances as we head into um like chapter nine on empirical data which is not super long followed by 10 which is just a restatement um onwards we go thanks y'all bye thanks everyone